

ARM-450 — Pre-Print Flight Check

Eight stages · 27 checks · run `python3 preflight.py`

[done] GO FOR PRINTING

21 passed · 0 failures · 2026-08-22

Every stage below is a real analysis, not a restatement of the one before, and each states its own criterion so a pass can be argued with. The verdict is GO only if every hard stage passes.

What each stage actually proves

stage	question it answers	criterion
1 Design check	Do the features physically exist in the geometry?	STEP B-rep audit, not parameters
2 FEA	Does anything yield under load?	p99 von Mises SF ≥ 3
	Does every load case hold with a tool fitted?	6 criteria × 5 configurations
3 Creep + fatigue	Does it fail slowly, in ways a static check misses?	stress ratio < 0.1, life > 1e8 cycles
4 Stiffness	Does the tool stay where it is put?	structural compliance < 0.30 mm
5 URDF	Does the model match the machine?	parses, arm mass matches the gate, meshes resolve, the fitted tool is present
6 Path tracing	Does the trajectory you will run actually work?	0 collisions, < 0.20 mm rms, inside limits
7 Simulation	Will the ROS side work on real hardware?	trajectory published, approach ramp, rate within servo
8 End effectors	Do the tools go on, and does the docking latch work?	29 interface checks incl. a simulated mate
	Does a fitted tool stay inside the agreed envelope?	system reach ≤ 495 mm

Results

STAGE 1 — DESIGN CHECK AND VERIFICATION

```
-----
[ PASS ] geometry audit (STEP B-rep)          52 features physically present, 0
missing/offset
[ PASS ] every bearing pocket accepts its bearing  10 pockets, light press
```

```

[ PASS ] pre-print gate                60 passed, 0 warnings, 0 failures
[ PASS ] mass within the hard ceiling    1365 g of 1450 g

STAGE 2 – FEA
-----
[ PASS ] FEA safety factor (p99 von Mises)    worst SF 23.0 across all parts

STAGE 3 – CREEP AND FATIGUE
-----
[ PASS ] working stress far below yield        0.38 MPa vs 60 MPa yield (ratio
0.0064)
[ PASS ] fatigue life                        4.3e+28 cycles to failure
[ PASS ] creep strain at 1 year, continuous load  0.040 % – park folded when idle

STAGE 4 – STIFFNESS
-----
[ PASS ] structural compliance at the tool      0.098 mm in PLA+CF at 2.0 mm wall
(design wall is 2.4)

STAGE 5 – URDF
-----
[ PASS ] URDF parses (mesh version)            single tree, root world
[ PASS ] URDF mass matches the mass gate        1.365 kg arm vs 1.365 kg (+ 75 g
tool as payload)
[ PASS ] the fitted tool is in the model RViz loads 75 g tool link on link6 – the
URDF had no tool at all until 2026-08-22
[ PASS ] every mesh reference resolves          8/8 meshes found

STAGE 6 – PATH TRACING
-----
[ PASS ] solved path is collision-free          240/240 waypoints clear (exact
mesh)
[ PASS ] path accuracy                        0.014 mm rms, tilt 0.093 deg rms
[ PASS ] path smoothness                      largest joint step 2.09 deg
between waypoints
[ PASS ] every waypoint inside the joint limits    checked against arm450.urdf,
including the narrowed J5

STAGE 7 – SIMULATION READINESS
-----
[ PASS ] node loads the solved trajectory        240 waypoints
[ PASS ] playback is continuous (ping-pong)      no wrap discontinuity
[ PASS ] approach ramp before tracing           eases from the start pose, no
slam on real servos
[ PASS ] JointTrajectory published for a controller 240 points, latched
TRANSIENT_LOCAL on /arm450/trajectory
[ PASS ] joint rate within the servo            0.91 rad/s vs ~2.3 rad/s loaded
ST3215

STAGE 8 – END EFFECTORS
-----
[ PASS ] end-effector interfaces                29/29 checks – J6 bolt pattern,

```

bayonet fit, rack mesh, docking capture

[PASS] heaviest tool within the payload allowance 75 g (adapter 14 + gripper 60 incl. SG90) vs the 300 g assumed at the TCP

[PASS] system reach with the longest tool fitted 492 mm (450 arm + 42 gripper) vs 495 mm accepted – the 450 mm spec is the ARM

Why stage 8 counts solids and simulates a mate

Stage 8 started as a bolt-pattern and bayonet-diameter check, and it passed while the docking tool had no latch on it. Three lugs had been unioned into the probe at a height where the bore was already Ø25.8, so they floated 5.5 mm clear of any wall. CadQuery's .union() does not fail when the shapes do not touch — it returns a compound — and every downstream step treats a compound as a part.

So the stage now asks two questions no dimension check can answer:

How many solids are you? len(shape.Solids()) for all six tool parts, fail on anything but 1. That one line would have caught the floating lugs, and it also catches the horn_adapter failure mode from earlier in the project.

Does the mate work? The exported probe mesh is driven onto the exported target mesh through all five states, with the expected answer written down for each:

state	expected
6 mm short, slots aligned	clear
seated	clear
rolled 30° to latch	clear
latched, pulled 2 mm	collides — this is the latch working
rolled back, withdrawn	clear

A check that only ever expects no collision cannot test a latch, because a latch is a deliberate collision. Capture tolerance is measured the same way — sweep the probe sideways until it fouls — rather than subtracted from two diameters: ±9.75 mm.

Why stage 6 and 7 exist separately

Stage 6 checks the path, not the space. An earlier version of this project ran a random-pose interference sweep, saw ~30 % collisions, and treated it as background noise because the joint-limit box is mostly irrelevant. The task was never checked — and when it finally was, 72 of 240 waypoints collided. Coverage of a state space is a much weaker claim than correctness of the one path that will be executed.

Stage 7 checks what hardware needs, which simulation does not expose. Three things were added specifically because they only bite on real servos:

- An approach ramp. The node used to start at waypoint 0. On real hardware the arm is

wherever it was left, so that is a full-speed slam. It now eases across on a cosine ramp over approach_time (default 3 s), leaving and arriving at zero velocity.

- A JointTrajectory topic. /joint_states is a visualisation stream — a real controller will not follow it. The whole solved path is now published once as a 240-point trajectory_msgs/JointTrajectory on /arm450/trajectory, latched (TRANSIENT_LOCAL) so a controller that starts later still receives it. Velocities are included by central difference.

- A velocity check against the actuator. Peak joint rate is 0.91 rad/s against roughly 2.3 rad/s for a loaded ST3215. Exceeding it would not error — the servo would simply lag, and the traced path would quietly stop being the solved path.

What GO does and does not mean

It means: the design is verified. Geometry, structure, kinematics, the model and the software all agree with each other and with the arm you are about to print.

It does not mean the print will be right. Two things remain, and neither is a design question:

1. The fit coupon. Every check above verifies the geometry is correct. None of them tells you what your machine does to a real Ø42 bore. That is a 0.9 h print and it gates every part with a bearing pocket.
2. Servo backlash is unmeasured and is the largest single term in the precision budget. The 0.014 mm rms above is what the mechanism allows; measured end-to-end precision is expected around 1.4 mm, dominated by backlash.

Running it

```
cd ~/ros2_ws/arm450_design
python3 preflight.py          # exits 0 on GO, 1 on NO-GO
```

Re-run it after any change to the CAD, the URDF, the trajectory or the node. It is cheap compared with a print run.